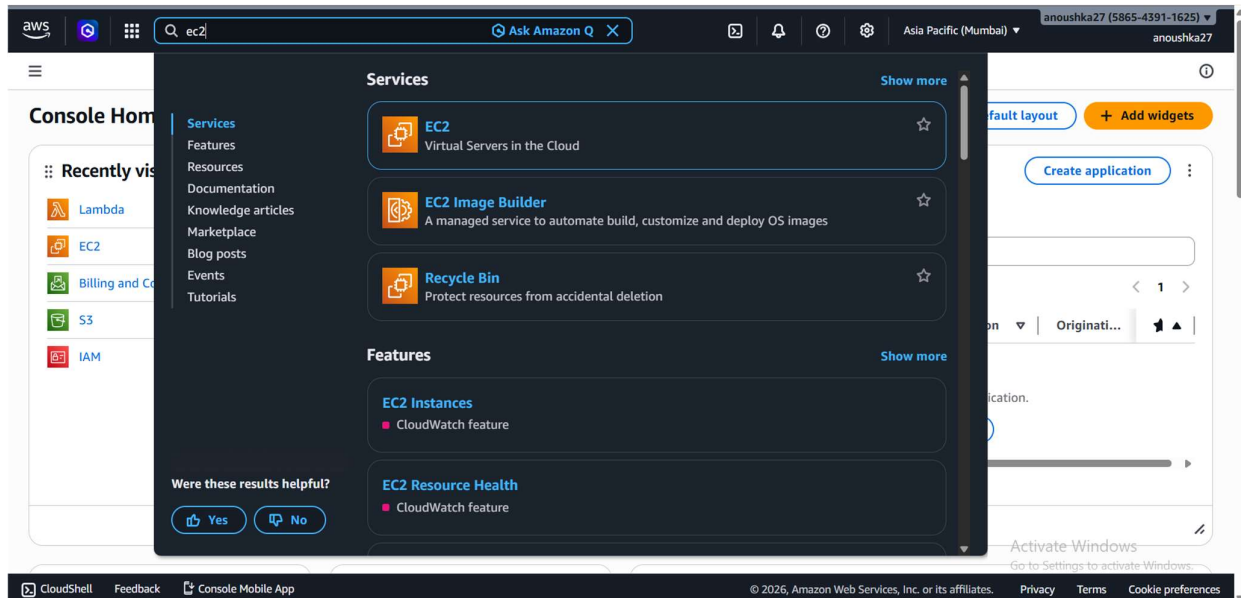


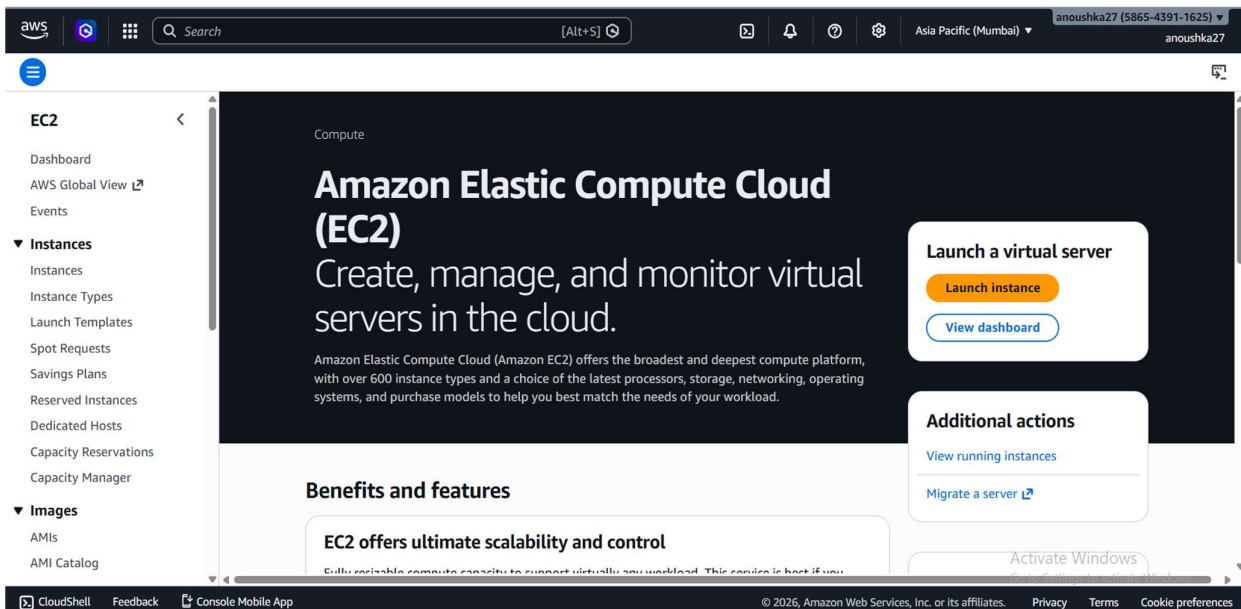
Assignment No. 14

Problem Definition: Create an Elastic IP for an Instance.

1) Sign up for an **AWS** account, search for '**EC2**' then click on it.



2) Select an **instance** or **launch** a new one and click **Launch Instance**.



3) Fill up the required details-> **'Name' (AnoushkaKoley)** and under **'Application and OS Images'** select **'ubuntu'**.

Name and tags [Info](#)

Name
AnoushkaKoley [Add additional tags](#)

Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Search our full catalog including 1000s of application and OS images

Recents **Quick Start**

Amazon Linux macOS **Ubuntu** Windows Red Hat SUSE Linux Debian

[Browse more AMIs](#)
Including AMIs from AWS, Marketplace and the Community

Summary

Number of instances [Info](#)
1

Software Image (AMI)
Canonical, Ubuntu, 24.04, amd64...[read more](#)
ami-05d2d839d4f73aafb

Virtual server type (instance type)
t3.micro

Firewall (security group)
Anoushka-group

Storage (volumes)
1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

Activate Windows
Go to Settings to activate Windows.

4) Select a key pair

Additional costs apply for AMIs with pre-installed software

Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*
keyyyy [Create new key pair](#)

Network settings [Info](#) [Edit](#)

Network [Info](#)
vpc-083fe2ad089e676d3

Subnet [Info](#)
No preference (Default subnet in any availability zone)

Auto-assign public IP [Info](#)
Enable

Firewall (security groups) [Info](#)

Summary

Number of instances [Info](#)
1

Software Image (AMI)
Canonical, Ubuntu, 24.04, amd64...[read more](#)
ami-05d2d839d4f73aafb

Virtual server type (instance type)
t3.micro

Firewall (security group)
Anoushka-group

Storage (volumes)
1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

Activate Windows
Go to Settings to activate Windows.

5) For '**Network Settings**' from '**Select existing security group**' choose the **security group** you created.

The screenshot shows the 'Launch an instance' page in the AWS Management Console. The 'Network Settings' section is active, showing options for Subnet, Auto-assign public IP, and Firewall (security groups). The 'Firewall (security groups)' section has 'Select existing security group' selected, and 'Anoushka-group' is chosen from the dropdown. The 'Common security groups' section shows 'Anoushka-group' selected. The 'Configure storage' section shows '1x 8 GiB gp3' storage. The 'Summary' panel on the right shows 'Number of instances' as 1, 'Software Image (AMI)' as Canonical, Ubuntu, 24.04, amd64, 'Virtual server type (instance type)' as t3.micro, 'Firewall (security group)' as Anoushka-group, and 'Storage (volumes)' as 1 volume(s) - 8 GiB. The 'Launch instance' button is visible.

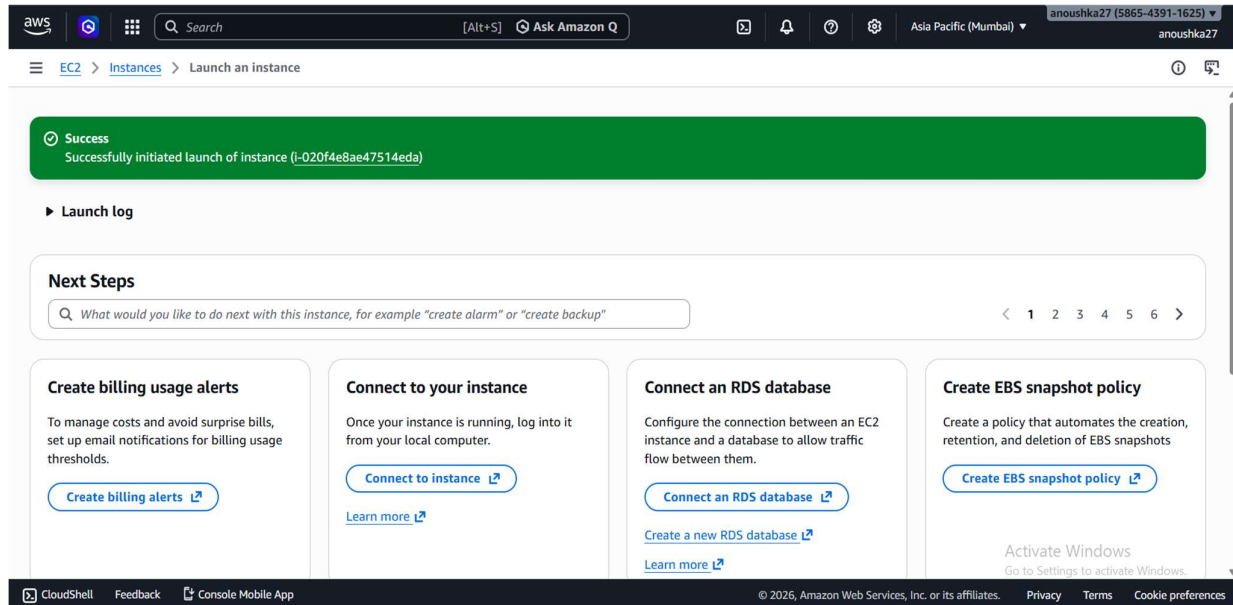
6) In '**Advanced details**' go to the last, in advanced data put the following bash script. Then click on '**Launch instance**'.

The screenshot shows the 'Launch an instance' page in the AWS Management Console, with the 'Advanced details' section expanded. The 'User data - optional' section is active, showing a text area with a bash script. The script is as follows:

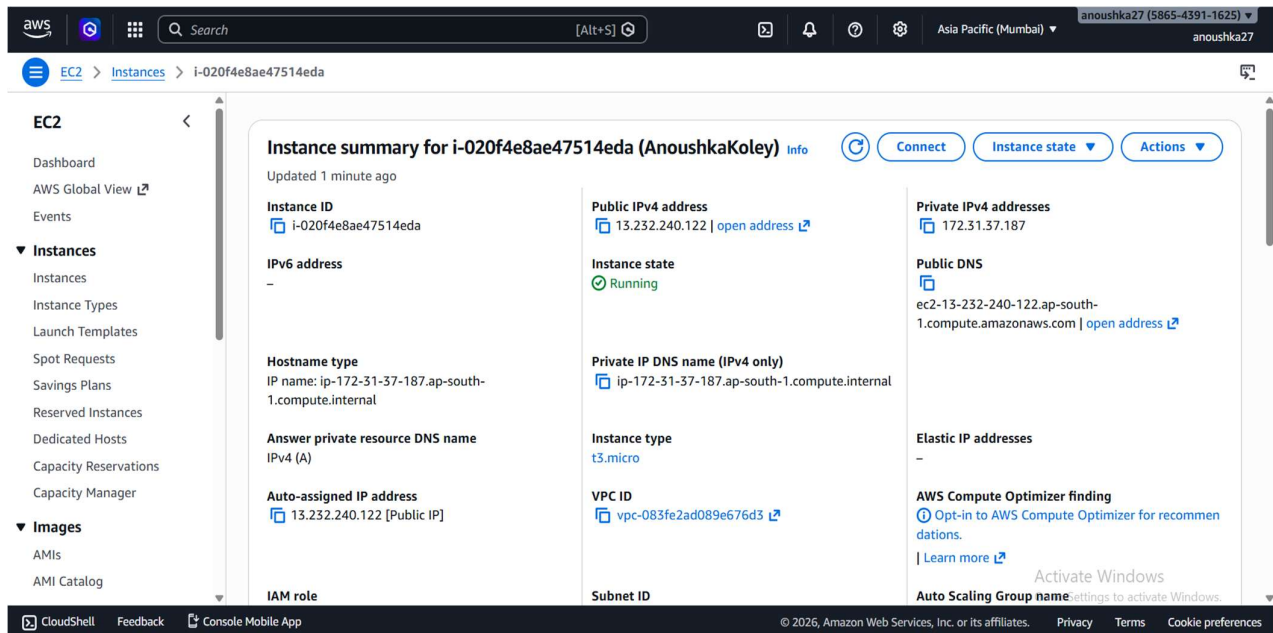
```
#!/bin/bash
apt-get update
apt-get install -y nginx
systemctl start nginx
systemctl enable nginx
apt-get install -y git
sudo curl -fsSL https://deb.nodesource.com/setup_18.x | sudo -E bash -
apt-get install -y nodejs
git clone https://github.com/Anoushka431/aws-repo.git
cd aws-repo
npm install
node index.js
```

The 'Summary' panel on the right shows the same configuration as the previous screenshot. The 'Launch instance' button is visible.

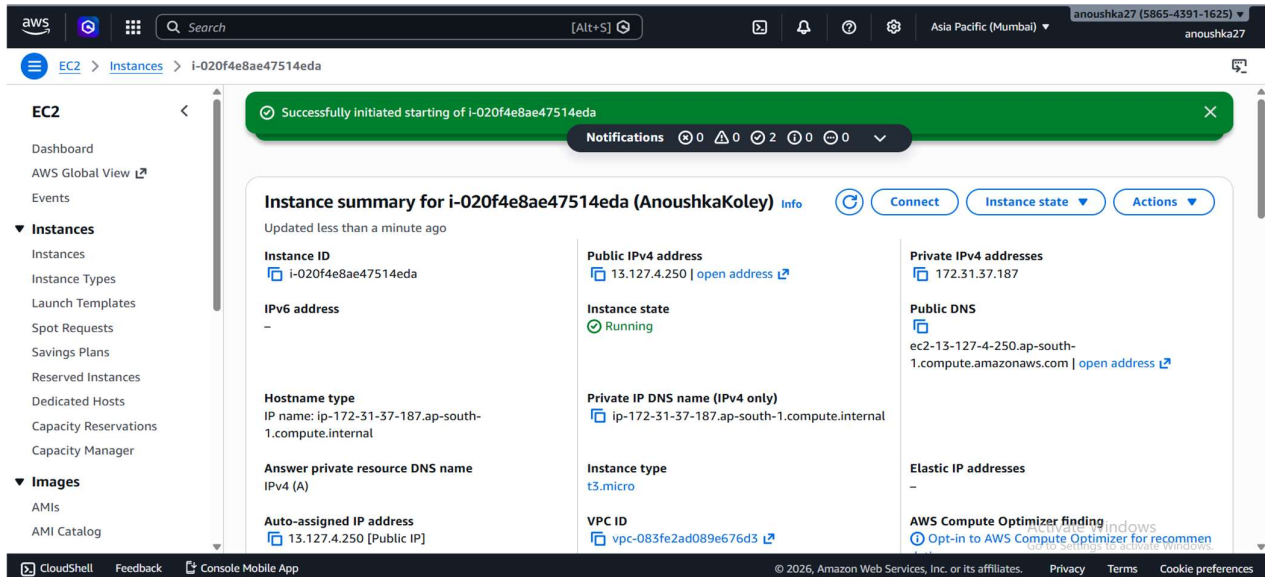
7) 'AnoushkaKoley' instance is created successfully.



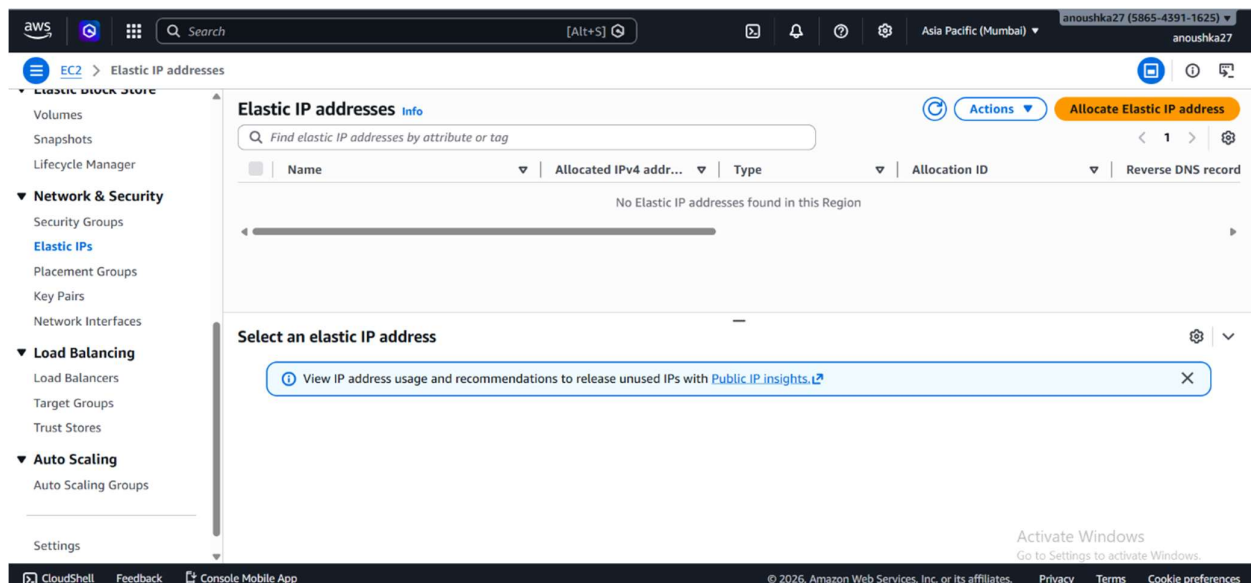
8) Save its **Public IPv4** address of this in a text file for future comparison.



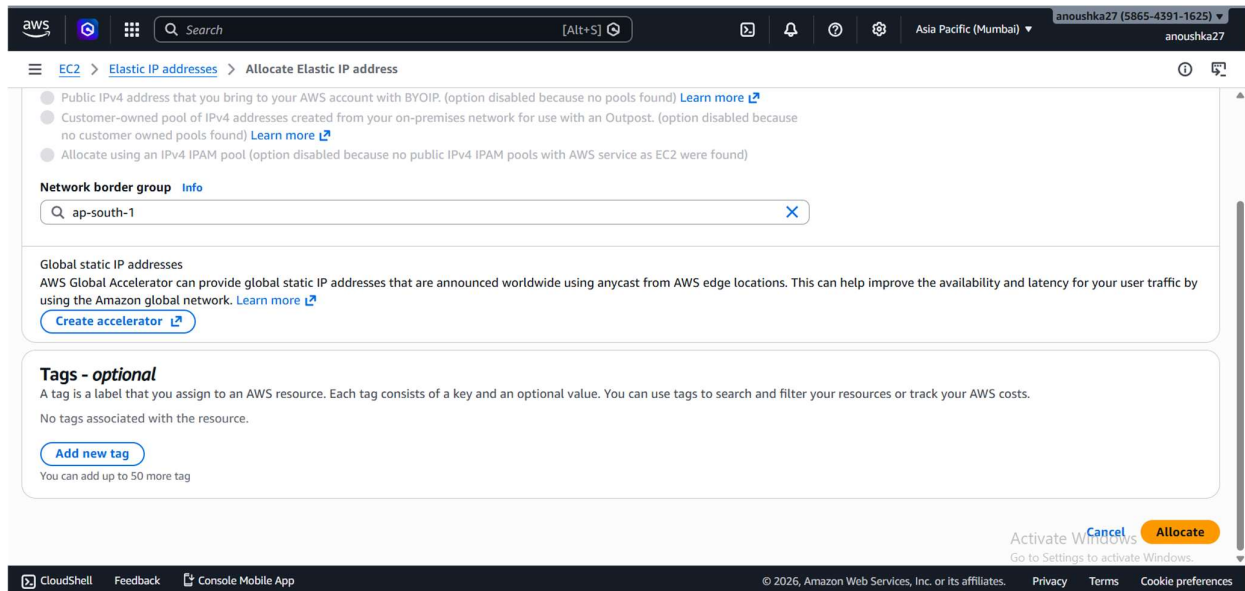
9) With the instance selected, open the **Instance State** dropdown menu and select '**Stop Instance**' option to temporarily put the instance in the sleep state. Now again **start** this instance.



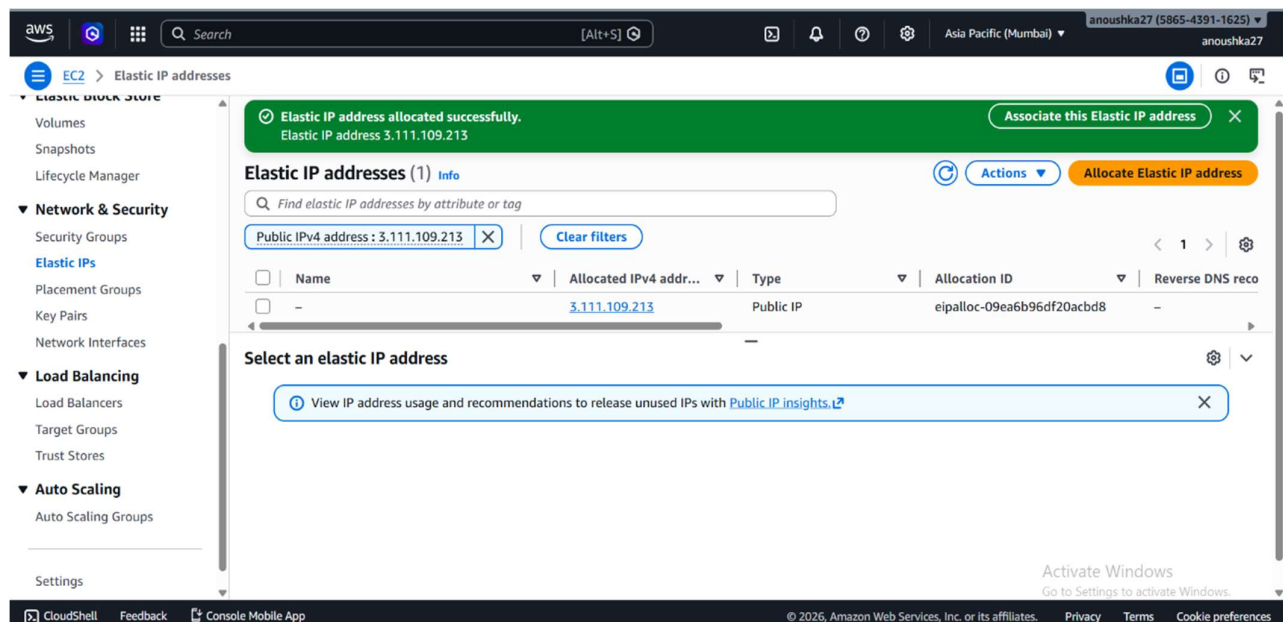
10) Now again save the **Public IPv4** address in the text file and compare it with the previous IP. It is evident that the IP address changes every time the instance is launched. To avoid it **Elastic IP** must be assigned. From the **EC2 Dashboard**, go to the '**Elastic Ips**' option. Then click on '**Allocate Elastic IP**'.



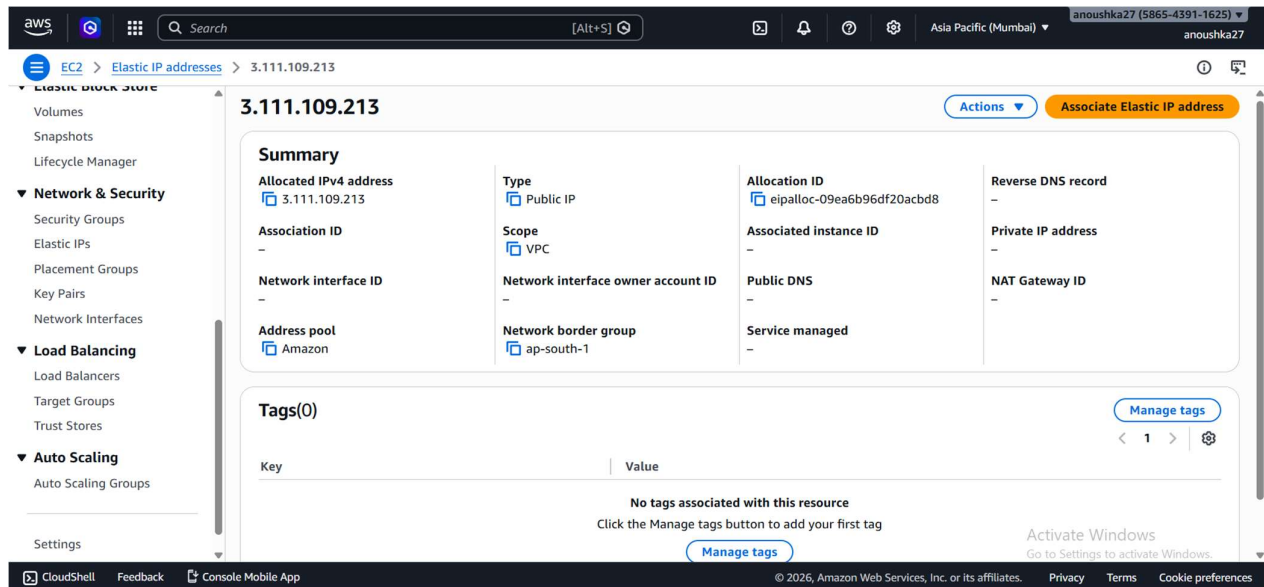
11) With 'Amazon's pool of IPv4 Addresses' selected under Public IPv4 Address pool, click on '**Allocate**'.



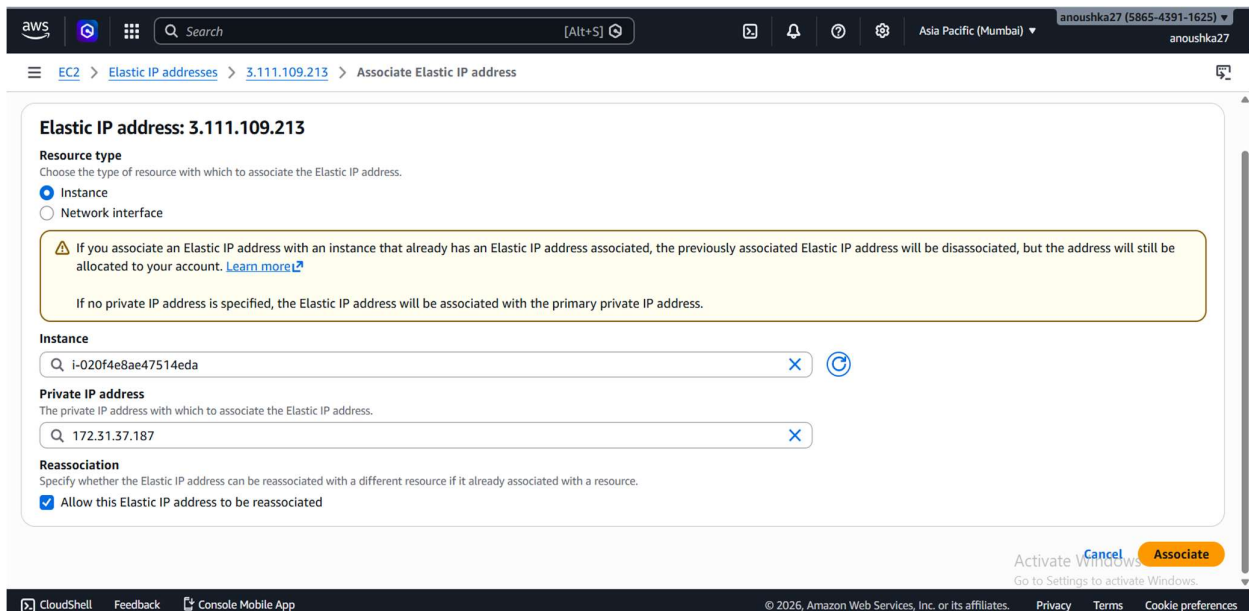
12) The **Elastic IP** has been created. Click on the **IP** to open it.



13) Click on '**Associate Elastic IP Address**'.



14) With resource type selected as 'Instance', select the name of the instance & **its private IP address**. Check the bottom checkbox to reassociate the Elastic IP. Then click on '**Associate**'.



15) **Elastic IP address** associated successfully.

The screenshot shows the AWS Management Console interface. At the top, a green notification bar states: "Elastic IP address associated successfully. Elastic IP address 3.111.109.213 has been associated with instance i-020f4e8ae47514eda". Below this, the page title is "3.111.109.213". The left sidebar shows the navigation menu with categories like Elastic Block Store, Network & Security, Load Balancing, and Auto Scaling. The main content area displays the "Summary" for the Elastic IP address. It includes fields for Allocated IPv4 address (3.111.109.213), Association ID (eipassoc-06e5abf8f797de600), Network interface ID (eni-060b10b8d745bbb1e), Address pool (Amazon), Type (Public IP), Scope (VPC), Network interface owner account ID (586543911625), Network border group (ap-south-1), Allocation ID (eipalloc-09ea6b96df20acbd8), Associated instance ID (i-020f4e8ae47514eda), Public DNS (ec2-3-111-109-213.ap-south-1.compute.amazonaws.com), Service managed (Reverse DNS record, Private IP address, NAT Gateway ID), and Tags (0). There are buttons for "Actions" and "Associate Elastic IP address". At the bottom, there are links for "CloudShell", "Feedback", and "Console Mobile App", along with copyright information for Amazon Web Services.

16) Copy the **IPv4 address** before selecting 'Stop Instance' and again **start** this instance and copy the **IPv4 address**.

The screenshot shows the AWS Management Console interface for the instance i-020f4e8ae47514eda. At the top, a green notification bar states: "Successfully initiated starting of i-020f4e8ae47514eda". Below this, the page title is "i-020f4e8ae47514eda". The left sidebar shows the navigation menu with categories like Instances, Images, and Elastic Block Store. The main content area displays the "Instance summary for i-020f4e8ae47514eda (AnoushkaKoley)". It includes fields for Instance ID (i-020f4e8ae47514eda), IPv6 address (None), Hostname type (IP name: ip-172-31-37-187.ap-south-1.compute.internal), Answer private resource DNS name (IPv4 (A)), Auto-assigned IP address (None), Public IPv4 address (3.111.109.213), Instance state (Running), Private IP DNS name (IPV4 only) (ip-172-31-37-187.ap-south-1.compute.internal), Instance type (t3.micro), VPC ID (vpc-083fe2ad089e676d3), Private IPv4 addresses (172.31.37.187), Public DNS (ec2-3-111-109-213.ap-south-1.compute.amazonaws.com), Elastic IP addresses (3.111.109.213 [Public IP]), and AWS Compute Optimizer finding (Opt-in to AWS Compute Optimizer for recommendations to reduce costs). There are buttons for "Connect", "Instance state", and "Actions". At the bottom, there are links for "CloudShell", "Feedback", and "Console Mobile App", along with copyright information for Amazon Web Services.

No changes are observed.